

DATA ANALYST REPORT

Wholesaler Cookies Sales Dataset

Wholesale Distribution Performance Analysis | Sep 2019 – Dec 2020

Data Source: V4 – Final Cleaned Dataset (700 orders × 8 fields)

Report Date: July 12, 2026

1. Business Problem

A wholesale cookie distributor sells six cookie varieties to five retail accounts. Leadership needs a clear, data-driven view of where revenue and profit are actually coming from in order to plan production, pricing, and account management for the next fiscal cycle.

Specifically, the business needs this analysis to answer:

- Which products and which retailer accounts generate the most revenue and profit?
- Are profit margins consistent across products, or do certain items quietly underperform?
- Is demand seasonal, and if so, when should production and inventory be scaled up?
- Which variables (units sold, revenue, cost, margin) move together — and which don't — so pricing and cost decisions rest on real relationships rather than assumption?

This report analyzes 700 cleaned wholesale orders placed between September 2019 and December 2020 to answer these questions and translate the findings into concrete, actionable recommendations.

2. Exploratory Data Analysis Insights

The analysis uses sheet V4, the final cleaned table in the workbook, containing 700 orders across 8 fields: Retailers_ID, Order_ID, Product Name, Units Sold, Date, Estimated Revenue, Estimated Cost Price, and Profit Margin. Key structural and business observations from exploration:

- **Complete, duplicate-free data:** all 700 rows have no missing values, and every Order_ID is unique. The raw extract (V1) originally had 6 fully blank rows, which were removed during cleaning (V1 → V2 → V4).
- **Six product lines, uneven mix:** Chocolate Chip accounts for 202 of 700 orders (29%) — nearly double any other flavor. White Chocolate Macadamia Nut and Sugar tie at 109 orders each; Oatmeal Raisin (94), Snickerdoodle (93) and Fortune Cookie (93) trail behind.
- **Five retailer accounts, moderate spread:** Retailer 3 is the largest account (206 orders, \$1.43M revenue), while Retailer 1 is the smallest (92 orders, \$521K revenue) — roughly a 2.7x gap between the top and bottom account.
- **Wide order-size variation:** Units Sold ranges from 200 to 4,493 per order (mean 1,608), and Estimated Revenue ranges from \$200 to \$23,985 per order (mean \$6,700) — a mix of small top-up orders and large bulk orders.
- **Margin is fixed per product, not negotiated per order:** each product carries one constant profit margin (e.g., Snickerdoodle 62.5%, Fortune Cookie 50%), so margin variation across the dataset is driven entirely by product mix, not by per-order discounting.
- **A recurring seasonal ordering pattern:** order counts double to 70 in October 2019, June 2020, October 2020 and December 2020, versus a steady baseline of 35 orders in most other months — consistent with bi-annual or holiday-driven bulk restocking.

3. Summary Statistics

Descriptive statistics for every numeric column in the cleaned dataset (V4, n = 700 orders):

Column	Count	Mean	Std Dev	Min	25%	50%	75%	Max
Retailers_ID	700	3.10	1.26	1	2.00	3.00	4.00	5
Order_ID	700	504,624.22	234,009.58	100,553	296,211.50	515,959.50	710,704.25	899,743
Units Sold	700	1,608.32	867.45	200	905.00	1,542.50	2,229.50	4,493
Estimated Revenue (\$)	700	6,700.36	4,623.79	200	2,854.00	5,871.50	9,580.00	23,985
Estimated Cost Price (\$)	700	2,818.84	1,968.46	100	1,234.38	2,423.60	3,925.70	10,993.13
Profit Margin (%)	700	57.30	3.84	50.00	54.17	58.33	60.00	62.50

Table 1. Count, mean, standard deviation, min, 25th/50th/75th percentiles, and max for all numeric columns.

Date Column

Column	Count	Min	25%	50% (Median)	75%	Max
Date	700	2019-09-01	2019-12-24	2020-05-16	2020-09-08	2020-12-01

Table 2. Order date range — mean order date falls on 2020-04-28, roughly the dataset's midpoint, confirming even coverage across the 16-month window.

Categorical Column — Product Name

Column	Count	Unique	Top (Mode)	Frequency
Product Name	700	6	Chocolate Chip	202

Table 3. Product Name is categorical (6 unique values); Chocolate Chip is the most frequent product, appearing in 202 of 700 orders.

Reading these together: order value is right-skewed (mean revenue of \$6,700 sits above the \$5,872 median), meaning a smaller number of large bulk orders pull the average upward. Profit Margin, by contrast, is tightly clustered (std dev of just 3.84 points on a 50–62.5% range) because it is fixed per product rather than varying order to order.

4. Data Filtering

To isolate the highest-impact segments of the business, the cleaned dataset was filtered along four practical business dimensions: order value, profit margin, retailer + product combination, and time period. Results:

Filter Applied	Orders	% of Total	Revenue Captured
Estimated Revenue > \$10,000	158	22.6%	\$2,135,919.50 (45.5% of all revenue)
Profit Margin ≥ 60%	295	42.1%	Chocolate Chip & Snickerdoodle orders only
Retailer 3 + Chocolate Chip	70	10.0%	\$589,200.00
Q4 2020 (Oct–Dec)	175	25.0%	\$1,071,527.00

Table 4. Key filtered views of the dataset used to isolate high-impact orders.

- **High-value orders (>\$10,000):** just 158 orders (22.6% of volume) generate 45.5% of total revenue — a small share of transactions drives nearly half the business, making these orders a priority for fulfillment reliability and account management.
- **High-margin orders (≥60% margin):** 295 orders (42.1%) come exclusively from Chocolate Chip and Snickerdoodle — the only two products crossing the 60% margin threshold — confirming these two SKUs are the most profitable per unit sold.
- **Retailer 3 × Chocolate Chip:** this single retailer-product combination alone contributes \$589,200, over 12% of total company revenue — a concentration worth monitoring for account-retention risk.
- **Q4 2020 (Oct–Dec):** 175 orders and \$1,071,527 in revenue (nearly 23% of the entire 16-month total) fell in just the final quarter, reinforcing the seasonal spike already seen in the monthly trend.

5. Correlation Insights

A Pearson correlation matrix was calculated across all numeric fields to understand which variables move together. The heatmap below visualizes the strength and direction of every pairwise relationship (deep red = strong positive, deep blue = strong negative, white = no relationship).

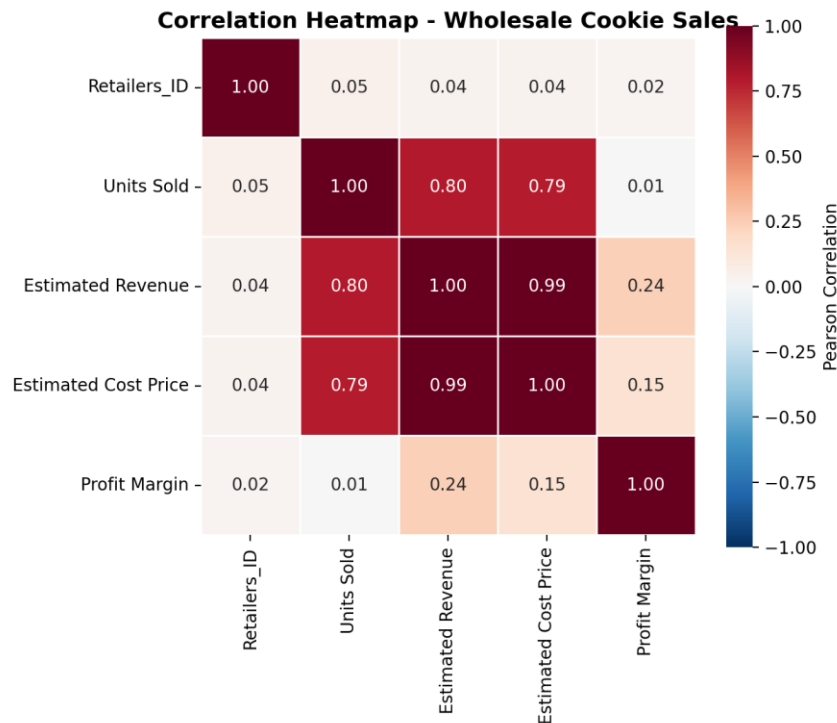


Figure 1. Pearson correlation heatmap across Retailers_ID, Units Sold, Estimated Revenue, Estimated Cost Price, and Profit Margin.

Column	Units Sold	Revenue	Cost Price	Profit Margin
Units Sold	1.00	0.80	0.79	0.01
Estimated Revenue	0.80	1.00	0.99	0.24
Estimated Cost Price	0.79	0.99	1.00	0.15
Profit Margin	0.01	0.24	0.15	1.00

Table 5. Full Pearson correlation coefficient matrix.

- **Revenue ↔ Cost Price: $r = 0.99$ (near-perfect positive).** This is expected — cost scales directly with order size, so larger orders cost proportionally more to fulfill.
- **Units Sold ↔ Revenue: $r = 0.80$, and Units Sold ↔ Cost Price: $r = 0.79$ (strong positive).** Order value is driven primarily by volume, not by price-per-unit differences between products.
- **Profit Margin ↔ Revenue: $r = 0.24$ (weak positive).** Higher-margin products tend to appear in somewhat larger orders, but the relationship is weak — margin is fundamentally a product attribute, not an order-size effect.
- **Profit Margin ↔ Units Sold: $r = 0.01$ (essentially none).** Confirms that how much of a product is ordered has no bearing on its margin — margin is fixed by product, not influenced by bulk purchasing.
- **Retailers_ID shows negligible correlation (≤ 0.05) with every other variable,** meaning no single retailer is systematically driving larger orders, higher revenue, or better margins — retailer identity and order economics are independent.

6. Research Questions & Key Findings

Q1: Which product drives the most revenue and profit?

Chocolate Chip leads decisively: 202 orders, 338,243 units sold, \$1,691,197.50 in revenue (36% of total company revenue), at a 60% profit margin — the highest revenue AND second-highest margin of any product. It is the clear flagship SKU.

Q2: Which product is the most profitable per order, and which underperforms?

Snickerdoodle carries the highest margin at 62.5%, though its lower order volume (93 orders, \$587,384 revenue) limits total profit contribution. Fortune Cookie is the weakest performer on every measure: the lowest margin (50%), the lowest total revenue (\$154,198), and the lowest average revenue per order (\$1,658) — roughly one-fifth of Chocolate Chip's \$8,372 average order value.

Q3: Which retailer account is most valuable, and how concentrated is the business?

Retailer 3 is the top account: 206 orders and \$1,431,191 in revenue (30.5% of total). Retailer 4 follows with \$1,108,643 (156 orders) and the highest average order value at \$7,106.70. Retailer 1 is both the smallest account (92 orders) and has the lowest average order value (\$5,665.80). The gap between the largest and smallest account is \$909,940 in annual revenue.

Q4: Is demand seasonal?

Yes. Ten of sixteen months show a steady baseline of 35 orders, but October 2019, June 2020, October 2020, and December 2020 each spike to 70 orders — exactly double. Q4 2020 alone (Oct–Dec) accounted for \$1,071,527, nearly 23% of all revenue in the 16-month dataset, confirming a strong year-end demand surge.

Q5: How do cost, revenue, and margin relate to one another?

Cost and revenue move almost in lockstep ($r = 0.99$) because cost is volume-driven. Profit margin, however, is essentially uncorrelated with order size ($r = 0.01$ with Units Sold) — margin is a fixed product-level attribute set at the SKU level, not something that emerges from bulk ordering or specific retailer negotiation.

Summary of Key Findings

- Chocolate Chip is the single largest revenue driver, contributing 36% of total company revenue at a strong 60% margin.
- Fortune Cookie is the weakest product on every metric — lowest margin, lowest revenue, and lowest per-order value.
- Retailer 3 is the most valuable account (30.5% of revenue); Retailer 1 is the smallest and has the lowest average order size.
- Just 22.6% of orders (those above \$10,000) generate 45.5% of total revenue — the business is meaningfully concentrated in large orders.
- Demand is seasonal, with order volume doubling in Oct 2019, Jun 2020, Oct 2020, and Dec 2020.
- Profit margin is fixed per product and does not respond to order size or retailer — it is purely a pricing/product decision.

7. Final Recommendation

Based on the findings above, the following actions are recommended:

- **1. Protect and grow the Chocolate Chip supply chain.** It drives over a third of all revenue at a healthy 60% margin — prioritize this SKU for inventory buffers and never let it stock out, especially heading into the Q4 seasonal spike.
- **2. Reassess or re-position Fortune Cookie.** With the lowest margin (50%), lowest revenue, and lowest per-order value, either renegotiate its cost structure to lift margin, bundle it with higher-margin products, or scale back production in its favor.
- **3. Deepen the Retailer 3 relationship while diversifying account risk.** Retailer 3 alone represents nearly a third of revenue; formalize this partnership (volume incentives, dedicated account support) while actively growing Retailer 1 and Retailer 5, the two smallest accounts, to reduce dependency on a single customer.
- **4. Plan production and staffing around the confirmed seasonal pattern.** Build inventory ahead of October and December, where order volume consistently doubles — treating these as planned peak periods rather than reactive surges.
- **5. Use order-value segmentation for account management.** The 22.6% of orders above \$10,000 already drive 45.5% of revenue — create a dedicated large-order fulfillment and account-service track to protect this concentrated revenue base from service failures.

- **6. Keep pricing and cost-control decisions separate from volume incentives.** Since profit margin is unrelated to order size, discounting for bulk purchases will not naturally protect margin — any volume discount strategy needs its own explicit margin floor.

Together, these actions focus resources on what the data shows are the highest-leverage points in the business: the best-selling product, the most valuable account, the predictable seasonal peaks, and the concentrated pool of large orders that already generate nearly half of all revenue.